

Lean WhatsApp Automation

Requirement, technology, cost, time and effort

Client: House Of Sandhya

Scope: Order placed -> day 2/day 4/day 6 WhatsApp if unshipped

Plus: Shipped tracking WhatsApp message

Hosting target: Cloudflare Workers + D1 + Cron

Development: Self-built with Codex assistance

This is a lean build estimate, not a full Shopify app or dashboard project.

1. Exact Requirement

Tumhari requirement sirf automated WhatsApp order updates hai. Koi full store app, dashboard, CRM, inbox, campaign builder ya marketing automation nahi banana.

Trigger / Time	Condition	Action
Order placed	Shopify/GoKwik order Shopify me create ho	Order data DB me save; 2/4/6 day jobs schedule
2 din baad	Order shipped/fulfilled nahi hua	WhatsApp processing update send
4 din baad	Order abhi bhi unshipped	WhatsApp quality/process update send
6 din baad	Order abhi bhi unshipped	WhatsApp dispatch-soon update send
Shipped hone par	Fulfillment/tracking available	WhatsApp tracking ID/link send

Important: Ye lean automation hai. Agar future me live courier events jaise in-transit/out-for-delivery/delivered chahiye, to Shiprocket/AfterShip integration alag phase me add hoga.

2. Technology Definitions

Cost batane se pehle har required technology ka meaning:

Thing	What it is	Why needed
GitHub repository	Code store karne ki jagah	Code backup, version control, Cloudflare deployment source
Cloudflare Worker	Serverless backend/API	Shopify webhook receive karega aur WhatsApp API call karega
Cloudflare D1	Small SQL database	Orders, scheduled jobs, message logs store karne ke liye
Cloudflare Cron Trigger	Scheduled runner	Har 30/60 minutes due jobs check karega
Shopify Webhooks	Shopify event notification system	Order create, order cancel, fulfillment create events receive karne ke liye
Shopify Admin API	Shopify order status read API	Message send karne se pehle latest fulfillment status check
Meta WhatsApp Cloud API	Official WhatsApp Business messaging API	Customer ko approved WhatsApp template messages bhejne ke liye
Meta message templates	Pre-approved WhatsApp message formats	Business-initiated order updates send karne ke liye mandatory

3. Accounts and Requirements

Requirement	Required?	Cost	Notes
GitHub account	Yes	INR 0	Private repo free plan enough
Cloudflare account	Yes	INR 0 to INR 420/month	Free may work; Paid Workers plan approx USD 5/month if needed
Meta Business Manager	Yes	INR 0	Business verification may be requested by Meta

Requirement	Required?	Cost	Notes
WhatsApp Business Account (WABA)	Yes	INR 0	Meta Business setup me create hoga
Dedicated WhatsApp API phone number	Yes	INR 0 if existing spare number; else SIM recharge approx INR 150-300/month	Normal WhatsApp app number directly use nahi karna best
Message template approval	Yes	INR 0	Meta approve karega; templates Utility category me hone chahiye
Meta billing/payment method	Yes	Usage based	Per delivered message charge
Domain name	No	INR 0	workers.dev URL enough hai; custom domain optional

4. Architecture

Step	What happens
1	Shopify order create webhook Cloudflare Worker URL ko call karega
2	Worker HMAC verify karega, order phone/order ID save karega
3	D1 me day 2, day 4, day 6 reminder jobs create honge
4	Cron Trigger periodically due jobs pick karega
5	Worker Shopify Admin API se latest fulfillment status check karega
6	Agar unfulfilled hai to WhatsApp Cloud API se Utility template send hoga
7	Fulfillment/shipped webhook aate hi tracking template send hoga
8	Message logs D1 me save honge: sent, skipped, failed, reason

Lean stack: Cloudflare Workers + D1 + Cron + Shopify Webhooks + Meta WhatsApp Cloud API.

5. Cost Estimate

Kyuki development hum khud karenge, developer/agency cost zero rakha gaya hai. Real cost mainly Cloudflare hosting aur WhatsApp delivered messages ka hoga.

Cost item	Definition	Expected cost
Development	Code hum yahan khud banayenge	INR 0
GitHub	Code repository	INR 0
Cloudflare Worker + Cron	Serverless backend + scheduler	INR 0 initially; safe budget INR 420/month if Workers Paid needed
Cloudflare D1 database	Orders/jobs/logs DB	INR 0 initially within free limits
Domain	Custom URL	INR 0; optional
Meta setup/templates	WABA, phone, template approval	INR 0 setup fee
WhatsApp messages	Meta charges delivered business messages by country/category	Planning budget INR 0.15/message for India Utility messages; exact live Meta pricing to confirm
Phone number	Dedicated WhatsApp API number	INR 0 if spare number; else INR 150-300/month SIM recharge

6. Monthly Cost Scenarios

Max 4 WhatsApp messages per order assumed: day 2, day 4, day 6, shipped tracking. Agar order jaldi ship ho gaya to actual messages kam honge.

Orders/month	Max msgs/order	Max msgs/month	WhatsApp @ INR 0.15/msg	Total with free Cloudflare	Total with paid Cloudflare
100	4	400	INR 60	INR 60	INR 480
500	4	2,000	INR 300	INR 300	INR 720
1,000	4	4,000	INR 600	INR 600	INR 1,020
3,000	4	12,000	INR 1,800	INR 1,800	INR 2,220
5,000	4	20,000	INR 3,000	INR 3,000	INR 3,420

Practical starting budget: INR 0-420/month fixed + WhatsApp usage. For most small scale cases, total monthly bill likely under INR 1,000 unless order volume grows.

7. Time and Effort

Phase	Work	Time	Effort
Planning	Template text, fields, flow final	0.5 day	Low
Meta setup	WABA, phone, templates, billing	1-3 days, may wait for approval	Medium
Core coding	Webhook, D1 schema, jobs, WhatsApp sender	2-3 days	Medium
Cloudflare deploy	Wrangler/GitHub deploy, env secrets, cron	0.5-1 day	Low-medium
Shopify setup	Webhooks/API access/secrets	0.5-1 day	Medium
Testing	Test orders, skipped/shipped/cancelled cases	1-2 days	Medium
Go-live	Final templates, live webhook, monitoring	0.5 day	Low

Total realistic time: 5-8 working days for lean automation, assuming Meta template approval is smooth. If Meta approval delays happen, add 2-5 days waiting time.

8. Final Total

Category	Amount
One-time development cost	INR 0, because we are building ourselves
One-time setup cost	INR 0, unless buying a new phone/SIM
Monthly fixed cost, lowest	INR 0 if Cloudflare free limits are enough
Monthly fixed cost, safe production budget	INR 420 Cloudflare Workers Paid + optional INR 150-300 phone recharge
Variable message cost	Approx INR 0.15 per delivered Utility WhatsApp message planning budget
Cost per order max	Approx INR 0.60/order if all 4 messages deliver
Recommended starting budget	Keep INR 1,000/month aside for Cloudflare + WhatsApp usage at small scale

9. What We Will Not Build Initially

To keep cost and effort low, phase 1 will not include:

Not included initially	Reason
Full Shopify embedded app UI	Not needed for message automation
Admin dashboard	Can be added later; logs in D1 are enough initially
Marketing/campaign messages	Higher compliance/cost; current need is order Utility updates
Live courier event tracking	Needs Shiprocket/AfterShip integration; add later if required
Multi-store support	Only House Of Sandhya needed now

10. Sources

1. Meta WhatsApp pricing: <https://developers.facebook.com/documentation/business-messaging/whatsapp/pricing>
2. WhatsApp Business Platform pricing: <https://whatsappbusiness.com/products/platform-pricing/>
3. Cloudflare Workers pricing: <https://developers.cloudflare.com/workers/platform/pricing/>
4. Cloudflare D1 pricing: <https://developers.cloudflare.com/d1/platform/pricing/>
5. Shopify webhooks: <https://shopify.dev/docs/api/webhooks/latest>